Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer with **2+ years of experience** building **Python**-driven data pipelines, telemetry-style analytics, and release-grade evaluation workflows across **AWS**, distributed systems, and autonomous-systems research.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Data Retrieval, Operating Systems

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built backend data infrastructure for a multi-tenant AI platform, processing **1K+ AI jobs** and **5K+ monthly integration jobs** while maintaining **99%+ reliability** across SQL-backed pipelines and cloud storage.
- Designed validation checks, job-state workflows, and performance instrumentation that reduced dashboard latency from **1.2s to 350ms** and improved recurring data-pipeline latency by **35%**.

USC AD Labs **Los Angeles**
ML Research Assistant *January 2025 – May 2025*
- Built simulation and model-training pipelines over **100K+ trajectory logs** and **10K autonomous-driving episodes**, converting HD-map and agent telemetry into measurable system signals for offline evaluation.
- Developed evaluation logic for long-horizon planning and closed-loop stability, improving planning performance by **18%** and system stability by **15%** in autonomous-driving experiments.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Implemented **Python/Flask** services and concurrent processing workflows for an AI grading platform handling **10K+ requests/day**, with secure ingestion, background execution, and operational monitoring.
- Tracked model usage, computed cost and rubric metrics, and supported AWS RDS-backed pipelines for **150+ instructors**, reducing feedback turnaround time by **60%**.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered AWS-based ETL and API workflows using **Glue**, **S3**, SQL, and CI/CD automation, improving response times by **30%** and increasing pipeline efficiency by **68%**.
- Investigated validation failures across automated build and test environments on Windows, Linux, and macOS, reducing technical debt by **20%** and improving release reliability.

Pandit Deendayal Energy University **India**
Data Analyst - Research Fellowship *May 2023 – August 2023*
- Built a detection pipeline over **650K+ URLs** with **21 behavioral indicators**, translating large-scale event data into statistical features and measurable model signals for anomaly classification.
- Achieved **96%+ detection accuracy** through comparative evaluation of ML and deep-learning models, using metrics and error analysis to refine the validation pipeline.

Skills

Languages: Python, SQL, Java, C++, JavaScript/TypeScript
Data & Metrics: Telemetry Analysis, KPI Computation, Statistical Analysis, Large Datasets, Time-Series/Validation Signals
Cloud & Pipelines: AWS (**S3**, **Lambda**, **ECS**, **Step Functions**, RDS, Glue), CI/CD, Workflow Orchestration
Systems: FastAPI, Flask, PostgreSQL, MySQL, Linux, Automated Testing, Logs/Failure Analysis, High-Reliability Systems

Projects

CaptionCraft - Image Caption Generator Live | *Transformers, PyTorch, Flask*
- Benchmarked captioning models on a **10K image / 100K caption dataset**, improving accuracy by **15%** (**81% to 96%**) while building inference APIs for scalable model evaluation.

Big-Brain Live | *FAISS, Pinecone, PostgreSQL, GCP*
- Engineered a cloud-based ingestion and indexing pipeline over **50K+ documents**, reducing retrieval latency by **40%** and supporting measurable retrieval-quality improvements through optimized data flows.

CPU-Scheduling Simulator Live | *C++, React.js, EJ-2*
- Implemented algorithmic scheduling evaluation across **1K+ simulated process combinations**, reducing debugging time by **20%** and improving average turnaround time by **25%**.